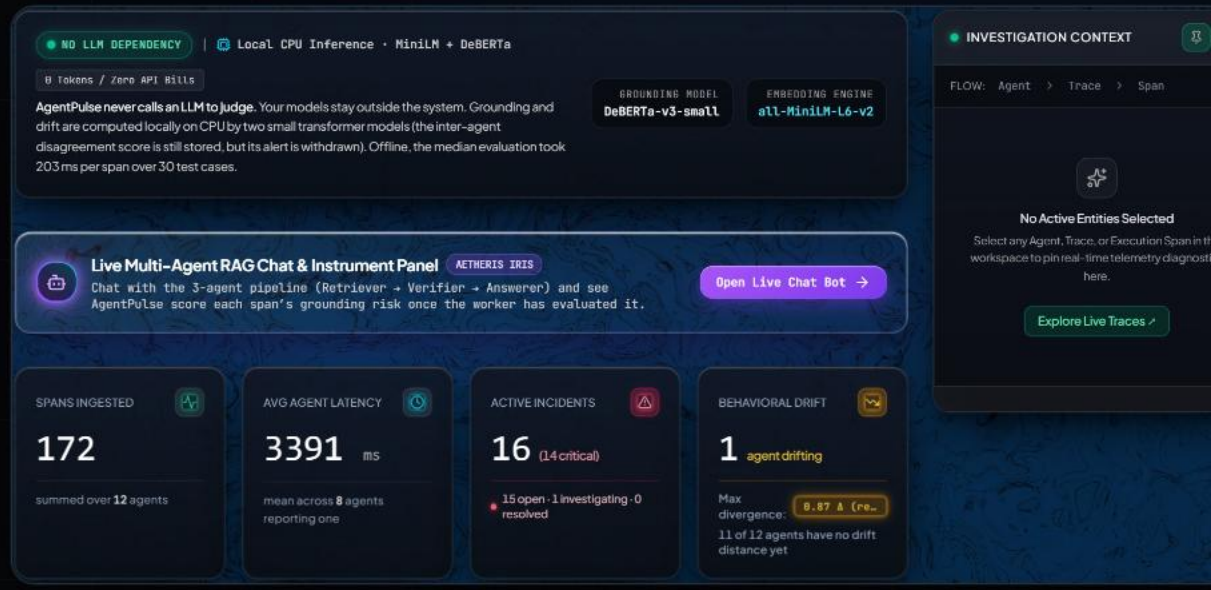
• M.TECH (AIML) · FINAL REVIEW · 3 NOV 2026

AgentPulse

Self-hosted observability for multi-agent LLM pipelines. Every span is scored for grounding risk and drift by two small local models — no LLM judge.



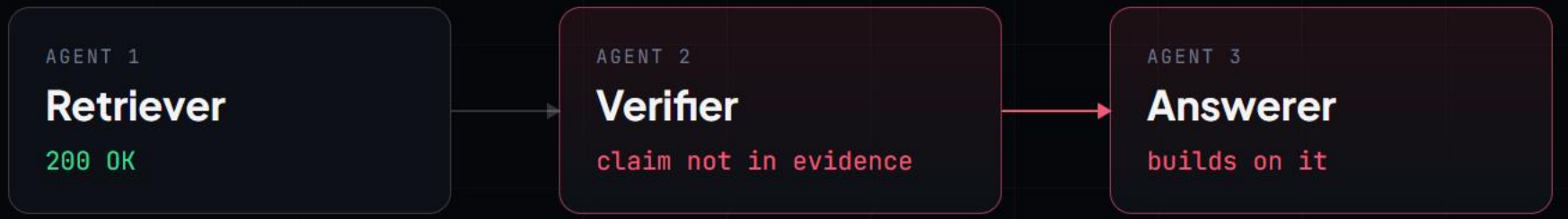
The live console, Overview (screenshot 1 Nov 2026)

Somnath Reddy

agentpulse.centralindia.cloudapp.azure.com
github.com/Soum-Code/agentpulse

• 01 · THE PROBLEM

Multi-agent pipelines fail *silently*



Grounding failure

An agent states something its own evidence does not support, and the next agent builds on it.

Behavioural drift

An agent's answers move away from how it used to answer, after a model or prompt change. Nothing errors.

Quality is an add-on

Existing tools trace tokens, latency and cost well, but treat quality evaluation as optional and sampled.

Goal of this project: score **every** span, cheaply enough for a CPU, and show plainly what was not measured. Framing from README.md (intro paragraph).

Score every span, *locally*

YOUR PROCESS
SDK
Decorator, OpenAI / Anthropic wrapper, LangGraph adapter.

`bounded buffer, fail-open`

→

FASTAPI
Ingest API
Accepts and queues. Runs no model.

`POST /v1/ingest`

→

STORAGE
SQLite (WAL)
Spans plus a durable job queue the worker leases from.

`at-least-once`

→

WORKER · CPU
Evaluator
MiniLM-L6-v2 embeddings + DeBERTa-v3-small NLI.

`grounding · drift`

→

RULES
Alerts
Thresholds per signal, one alert per agent and type in 15 min.

`cooldown 900 s`

→

REACT 19
Console
REST + WebSocket. Traces, incidents, drift, Lab.

`unmeasured = "—"`

202.94 ms

median per case, AgentPulse evaluator, vs **3,170.39 ms** for an LLM judge (Qwen3-8B, CPU), 30 cases

0.005 ms

SDK decorator overhead per call (P50 over 2,000 calls)

0

LLM calls in the evaluator: two small models, run on the CPU of the same VM

Sources: LLM_JUDGE_COMPARISON_REPORT.md (median latency table, 2026-08-26) · benchmarks/benchmark_results.json (2026-08-18) · HOW_IT_WORKS.md (buffer 10,000 spans, drop-oldest; cooldown 900 s) · README.md "Architecture in short"

Grounding risk: an NLI model, *not a judge*

```
risk = P(contradiction) + 0.3 × P(neutral)
```

DeBERTa-v3-small compares an agent's output with the evidence it was given. 0 = safe, 1 = risky. **High above 0.7, medium above 0.4.**

`contradicted` `unverifiable` `supported`

The verdict is whichever probability is highest. **Unverifiable** = the evidence neither supports nor contradicts. A status, not a pass.

0.3 is a design choice, not a fit: weights 0.0 to 0.3 cannot be told apart on this data.

0.963

F1 on the 30-case test split (FPR 0.059: 1 of 17 supported cases)

0.927 / 0.952

F1 on 47 truly held-out cases (at 0.7 / at 0.4), FPR 0.074

68

hand-labelled cases behind these numbers (39 supported, 29 failures). 5 of the 30 test cases copy dev cases. Small sample, not production traffic.

Source: GROUNDING_RECALIBRATION_2026-10.md (tables T3c, T3d, section 3.6; analysis 2026-10-22) · HOW_IT_WORKS.md (verdict rule) · backend/app/services/grounding.py (NEUTRAL_RISK_WEIGHT = 0.3)

Drift: how far an agent has moved *from itself*



schematic, not data

MiniLM embeds each output. Per agent, the mean of its **first 20** outputs is the baseline; the mean of its **latest 12** is the window. `DRIFT_DETECTED` fires above a cosine distance of **0.30**.

1.5%
false alarms

91.9%
detection

24.5%
of real sessions long enough to measure at all

Offline, an external real-trace corpus, held-out split, measured once.

Source: DRIFT_REAL_TEXT_DIAGNOSIS_REPORT.md sections 11.3 and 11.5 (2026-08-27) · README.md "What it does" · HOW_IT_WORKS.md (baseline 20, window 12, threshold 0.30)

• 05 · RESULTS · COST

Cheap enough to run on *every* span

MEDIAN EVALUATION TIME PER CASE, 30 CASES, CPU



One judge model on one machine: not a claim about every LLM judge.

180 ms

live median evaluation time over 172 spans on the deployed worker (ONNX), as the Settings view showed it on 27 Oct 2026

SINGLE RUNS, ONE WINDOWS 11 MACHINE, CPU, PYTORCH

SDK decorator overhead (P50, 2,000 calls)	0.005 ms
MiniLM embedding (P50, 100 runs)	15.1 ms
DeBERTa NLI (P50, 50 runs)	88.5 ms
Two grounding models, per case (30 cases)	151.19 ms
Full pipeline, per case (30 cases)	165.18 ms

Sources: LLM_JUDGE_COMPARISON_REPORT.md (2026-08-26) · benchmarks/benchmark_results.json (2026-08-18) · experiments/results/ablation_results.json (2026-09-18) · docs/E2E_TEST_2026-10-27.md item 23 (180 ms). Each is one run on one machine, not a constant.

• 06 · RESULTS · VERIFICATION

Tested, and checked *live*

628

backend tests passed (pytest, 4 min 12 s)

404

dashboard tests passed (35 files), tsc clean

Re-run 27 Oct 2026, author's machine

END-TO-END TEST OF THE LIVE SITE, 27 OCT 2026, FROM OUTSIDE

- ✓ **Health**: /v1/health, /v1/health/ready, /chat/health all 200
- ✓ **Closed routes**: /v1/keys and 3 path variants, ingest without key, PUT /v1/simulate: all 401
- ✓ **Chat**: one turn 6.35 s, all three agents answered by Groq; the worker scored all three spans
- ✓ **Telemetry Lab**: hallucination scenario raised HIGH alerts for verifier (0.971) and analyst (0.999)
- ✓ **Console**: every view in Dark and Chalk scripted for NaN / undefined / null: 0 found
- ! **Two real bugs found and fixed** (SVG arrowhead, header at 1366 px); one noted (stale worker count)

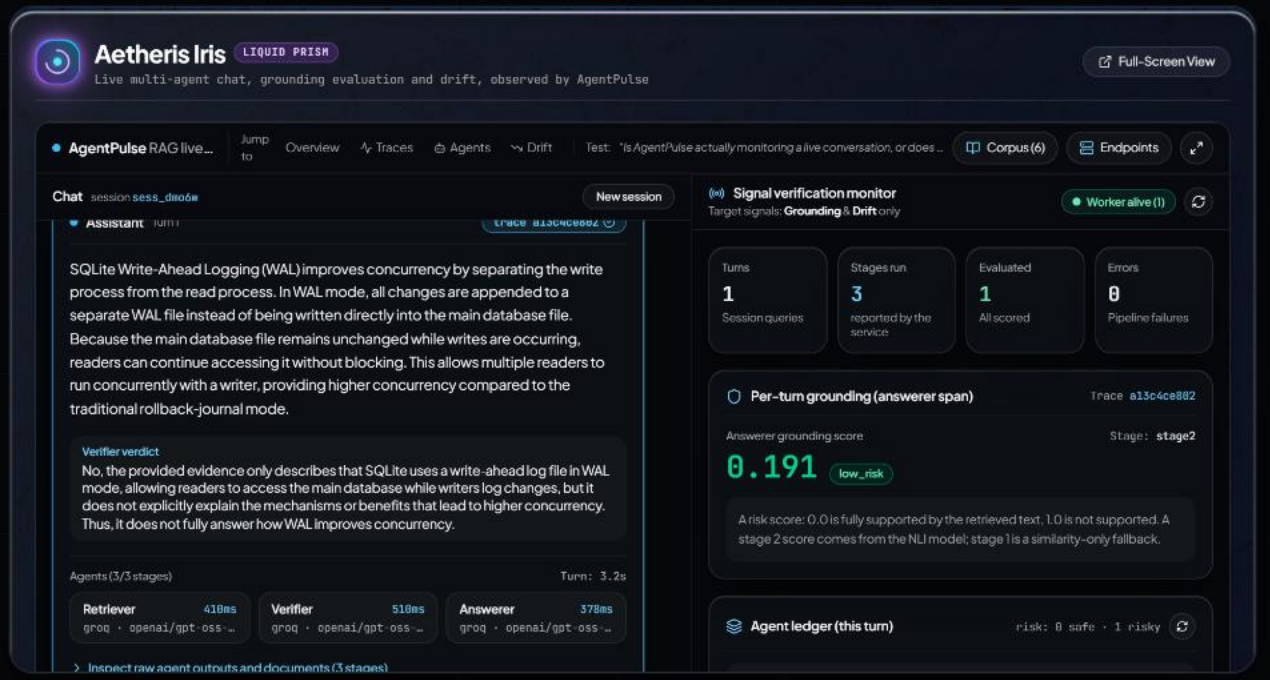
Sources: README.md "Testing and benchmarks" (628 / 404, re-run 2026-10-27) · docs/E2E_TEST_2026-10-27.md items 1-25 and findings F1-F3 · SESSION_HANDOFF.md 29.19

• 07 · LIVE DEMO

See it on the *live* site

https://agentpulse.centralindia.cloudapp.azure.com

1. **Launch App → RAG Live.** Ask "how does SQLite WAL improve concurrency?"
2. **Three stages**: retriever, verifier, answerer, each with time, provider, model.
3. **Agent ledger**: grounding risk and verdict per span; what "Unverifiable" means.
4. **Out of corpus**: "who won the 2022 football world cup?" Read the ledger.
5. **Traces** → newest trace → X exports the JSON.



A real turn, 1 Nov 2026: 3.2 s, all three agents on Groq; answerer grounding 0.191 (low risk). One turn, not a rate.

• 08 · ENGINEERING

Deployed, automated, *locked down*

DEPLOYMENT

One Azure VM

Standard_B2als_v2 (2 vCPU, 4 GB), Central India. Docker Compose: Caddy (TLS), nginx + dashboard, FastAPI backend, evaluator worker, chat service.

Nightly database backup.

CI / CD

GitHub Actions

Pull requests run the full pytest suite and the dashboard's tsc + vitest.

A push to main deploys: rebuild changed images, test the new backend on a copy of the live database, swap, roll back if unhealthy.

SECURITY

Closed by default

Every write needs the Caddy write password, except the chat and `POST /v1/simulate`. Key routes: 401 on every case variant. Refuses to start with the placeholder key.

Rate limits per IP: chat 5/min, 40/day; Lab 6/hour, 30/day; 300/day per site.

Sources: deploy/azure/README.md (VM size) · .github/workflows/tests.yml, dashboard.yml, deploy-azure.yml · deploy/azure/Caddyfile · README.md "What works live" · SESSION_HANDOFF.md 29.17.2 and the #109 entry in 29.18

09 · HONEST LIMITS

What did not hold up, *said first*

Inter-agent disagreement

Found **0 of 10** labelled contradictions on real traces (AUC 0.457 without the planner). Alert withdrawn; score only stored.

Tool-claim check

F1 0.000 on real traces: no claim found in 8,353 prose spans. It only reads counts written in digits.

Small labelled set

Grounding quality rests on **68** hand-labelled cases; only one is a neutral failure, so the data cannot say what the 0.3 weight buys.

Drift coverage

Only **24.5%** of real sessions were long enough to report a drift distance at all.

One small deployment

One VM, one SQLite file, one evaluator worker; backup on the same disk; deploy not gated on the test suite; the chat runs on free LLM tiers and can be slow or refused.

Sources: README.md "Known limits" · COMPETITIVE_POSITIONING.md 5.2 · SESSION_HANDOFF.md 24.9.5 · TOOL_CLAIM_EXTERNAL_TEST_REPORT.md section 10 · GROUNDING_RECALIBRATION_2026-10.md · DEMO_GUIDE.md "The honest limits"

Next, in order of what the limits *ask for*

GROUNDING

More labelled data

A larger labelled set drawn from real traces, with neutral failures, then re-run the ablation at the 0.3 weight.

THE TWO FAILED SIGNALS

Redesign or remove

Claim extraction from prose instead of digits for tool-claim; a different formulation for disagreement, tested on real traces first.

SCALE AND SAFETY

Beyond one VM

A server database instead of one SQLite file, several workers, an off-VM backup, deploys gated on the test suite.

REACH

More frameworks

LangChain and CrewAI adapters (today: decorator, OpenAI / Anthropic wrapper, LangGraph).

• CONCLUSION

Two signals that held up, *measured, live, and honest* about the rest.

01

Every span scored by two local models on a CPU. No LLM judge: 202.94 ms vs 3,170.39 ms median per case.

02

Grounding and drift held up. F1 0.963 on the test split; two other signals failed on real traces and are labelled so.

03

Live on Azure and tested. 628 + 404 tests; chat, Telemetry Lab and console open to examiners.

LIVE SITE

agentpulse.centralindia.cloudapp.azure.com

github.com/Soum-Code/agentpulse

Thank you · Questions?

BACKUP B1 · Q&A

The label-order bug, and why the weight is 0.3

Before 21 Oct the code read DeBERTa's **entailment and neutral probabilities swapped**. PR #69 reads the label order from the model config; every grounding figure was re-measured.

At the 0.7 cut-off, weights up to 0.6 give identical results: **high alerts do not depend on the weight**. Only the medium line does.

Live rows re-scored at 0.3: `90 low / 7 medium / 19 high` (at 0.5: 12 / 81 / 23).

TEST SPLIT, CUT-OFF 0.4, CORRECTED LABELS

weight	F1	false-positive rate
0.0 – 0.3	0.963	0.059
0.4	0.897	0.176
0.5	0.788	0.412

At 0.5, 7 of 17 supported test cases were flagged medium.

Source: GROUNDING_RECALIBRATION_2026-10.md (banner, findings 4-7, T3c, T5b, T7) · SESSION_HANDOFF.md 29.11.3 · PR #69, PR #89

BACKUP B2 · Q&A

Two signals measured, and withdrawn

TOOL-CLAIM VALIDATION

0.842 → 0.000

F1 on 19 hand-written cases → F1 on a real-data benchmark

Compares a count the output states with the tool's recorded result count. On real agent traces it extracted no claim from 8,353 prose spans: agents do not write counts in digits.

INTER-AGENT DISAGREEMENT

0 / 10

labelled contradictions found on real multi-agent traces

NLI between agents' outputs in one trace. With the planner excluded, AUC 0.457: chance. The alert was withdrawn; the score is still stored, not alerted on.

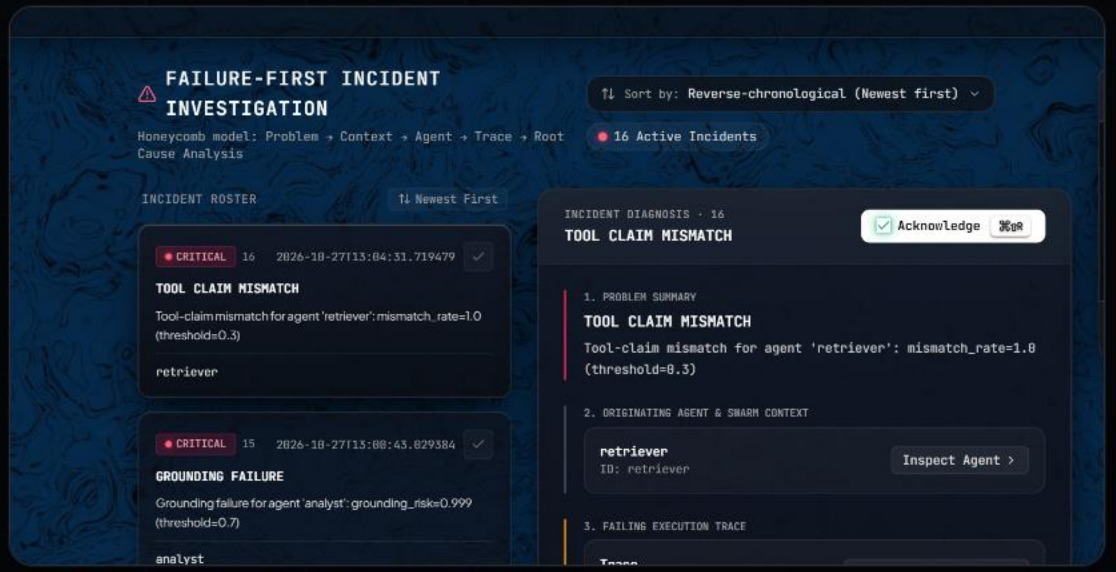
Sources: TOOL_CLAIM_EXTERNAL_TEST_REPORT.md section 10 · COMPETITIVE_POSITIONING.md 5.1, 5.2 · SESSION_HANDOFF.md 14.1, 24.9.5 · README.md "What it does"

BACKUP B3 · Q&A

Telemetry Lab: a fault in, an incident out



Five scenarios: clean, hallucination, tool-claim mismatch, drift, disagreement. Agent text is a fixture; the evaluator is real.



Incidents from Lab runs on 27 Oct: GROUNDING_FAILURE and HIGH_HALLUCINATION_RISK for analyst, risk 0.999.

Sources: HOW_IT_WORKS.md (Telemetry Lab; fixture text in routers/ingest.py) · docs/E2E_TEST_2026-10-27.md items 11, 12, 22 · README.md (Lab limits 6/h, 30/day per IP)